

3.1

Laurent Series

If $f(z)$ is an analytic function in a ring shaped region R bounded by two concentric circles C_1 and C_2 of radii r_1 and r_2 ($r_1 > r_2$) with centre at a , then for all z in R ,

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + \frac{b_1}{z-a} + \frac{b_2}{(z-a)^2} + \dots \quad \text{--- (1)}$$

$$\text{where } a_n = \frac{1}{2\pi i} \int_{C_1} \frac{f(w) dw}{(w-a)^{n+1}}$$

$$\text{and } b_n = \frac{1}{2\pi i} \int_{C_2} \frac{f(w) dw}{(w-a)^{-n+1}},$$

where the integrals are taken around C_1 and C_2 in the anticlockwise sense.

The RHS of (1) is called the Laurent's Series expansion of $f(z)$ about $z=a$ (or, around $z=a$).

Examples

① Find the Laurent's series for $f(z) = \frac{1}{(z-2)(1-z)}$

in the annulus region $1 < |z| < 2$.

Sol: Here $|z|=1$ and $|z|=2$ are two concentric circles with centre at 0 and radii 1 and 2 respectively. Clearly $f(z) = \frac{1}{(1-z)(z-2)}$ is

analytic in the region $1 < |z| < 2$.

$$\text{Now, } f(z) = \frac{1}{(1-z)(z-2)}$$

$$= \frac{-1}{(z-1)(z-2)}$$

$$= \frac{1}{z-1} - \frac{1}{z-2}$$

$$= \frac{1}{z(1-\frac{1}{z})} + \frac{1}{2(1-\frac{z}{2})}$$

$$= \frac{1}{z} \left(1 - \frac{1}{z}\right)^{-1} + \frac{1}{2} \left(1 - \frac{z}{2}\right)^{-1}$$

$$= \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots\right) + \frac{1}{2} \left(1 + \frac{z}{2} + \left(\frac{z}{2}\right)^2 + \dots\right)$$

valid for $1 < |z| < 2$.

② Expand $f(z) = \frac{1}{z^2 - 3z + 2}$ as a Laurent's series
in the region $0 < |z - 1| < 1$.

③ Find the Laurent series for $f(z) = \frac{7z - 2}{(z + 1)z(z - 2)}$

in the region $1 < |z + 1| < 3$.